

Rohan Singh

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EDUCATION

University of Virginia

Charlottesville, VA

BS in Computer Science – Minor in Data Science

May 2027

- Relevant Coursework: Machine Learning, Software Engineering, Computer Architecture, Data Structures, Algorithms

EXPERIENCE

Expedia Group

Jun 2026 – Present

Machine Learning Engineer Intern / Python, FastAPI, Trino SQL

San Jose, CA

- Built a real-time incentive decisioning service with **FastAPI** for the **98.6M-member** OneKey loyalty program.
- Proved a **+0.58pp causal booking lift** from a \$15 incentive via randomized holdout analysis of **1M+ members**.
- Showed **94%** of incentivized bookers were windfall using **T-learner CATE**, making the case for uplift targeting.
- Distilled **112 features to 27 servable** for a **two-tower reward recommender** via a train/serve feature contract.

Freddie Mac

May 2025 – Jul 2025

Software Engineer Intern / Java, TypeScript, Spring Boot, Angular

McLean, VA

- Built **Java Spring Boot** microservices that process and validate 120k accounting events, deploying via **Jenkins** CI/CD.
- Designed **GraphQL** schema and resolvers to consolidate 200+ REST endpoints, cutting **Angular** client payload by 60%.
- Led POC comparing **Playwright** and **Cypress**, achieving **4x faster** tests from 8 to 2mins and driving tool migration.
- Automated nightly data refresh via **Python + Jenkins**, moving datasets to UAT and cutting load time 2hrs to 10mins.
- Applied business rules to classify transactions, ingesting data from **AWS S3** and storing results in **MongoDB**.

Biocomplexity Institute

Nov 2024 – May 2025

AI/ML Research Assistant / Python, Slurm

Charlottesville, VA

- Built scalable graph-analysis pipelines with **cuGraph**, **SciPy**, and **NetworkX** on 1.2M-edge ICIJ and crypto graphs.
- Raised AUC-ROC from 0.55 (logistic baseline) to **0.76**, surfacing **250 anomalies** at **89%** precision for future auditors.
- Applied advanced centrality metrics and **One-Class SVM** to flag and rank illicit transaction patterns in networks.
- Optimized GPU-parallel workflows via **cuGraph**, scaling end-to-end anomaly detection to **2M+ nodes and edges**.
- Validated model on IBM **AMLSim** sandbox, confirming robust performance across real and synthetic financial graphs.

Cubic Corporation

Jun 2024 – Aug 2024

Software Engineer Intern / JavaScript, React, Terraform

Sterling, VA

- Built and deployed an AWS-hosted **Docusaurus** portal with **React** and **Terraform**, serving **500+** engineers.
- Integrated **AWS CodeBuild** to run lint and test checks on pull requests, reducing manual QA and expediting merges.
- Cut onboarding tickets **85%** (90 to 13) in one month by redesigning site flow and UI after user testing.
- Added **Algolia DocSearch** and simplified navigation; 95% top-3 result rate and median query latency 120ms.

PROJECTS

Marathon Prep Bot | TypeScript, Node.js, Garmin FIT SDK

Jun 2026 – Present

- Engineered a **TypeScript** pipeline decoding binary **Garmin FIT** files into **1,300+** CRC-validated activity records.
- Implemented **Banister TRIMP**, **VDOT**, and **aerobic decoupling** models over per-second HR streams to quantify load.
- Built an **LLM-agent** coaching loop turning **13 months** of training data into weekly plans on a live dashboard.
- Automated a 6-hour iCloud-to-production sync with a strict **field whitelist** keeping GPS and health data local.
- Wrote **202 regression tests** across 21 suites in strict TypeScript, covering FIT decode round-trips and load models.

Personalized Movie Ranking System | Go, Python, LightGBM, Next.js

Dec 2025 – Jan 2026

- Built a production-style movie ranking system with offline ML training and a low-latency **Go** API for ranking queries.
- Ingested and validated the **MovieLens 32M** dataset (200,948 users, 87,585 movies) and enriched items via **TMDB**.
- Trained a **LightGBM ranker** with time-based splits, improving **NDCG@10 by 11.8%** vs baseline on **3.20M rows**.
- Implemented a **Go HTTP service** ranking movies from precomputed features with **16.6ms p95 latency**.
- Built a lightweight **Next.js + Tailwind** demo UI with movie search and ranked result cards for demos.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Go, C++, SQL

Frameworks & Libraries: React, Next.js, Angular, Spring Boot, Flask, GraphQL, LightGBM, scikit-learn, SciPy, NetworkX, cuGraph, REST, Tailwind CSS

Cloud & DevOps: AWS, Azure, Docker, Jenkins, Gradle, Terraform, Git, Cypress, Playwright, Cucumber

Databases: MongoDB, PostgreSQL, MySQL

Certifications: AWS Cloud Practitioner, Oracle Cloud Infrastructure Generative AI Professional, PCAP